Project Titles

1. AI-Based Intelligent Network Design Assistant for Automated Enterprise Network Planning
2. AI-Based TCP Congestion Window Prediction Using Deep Learning and Reinforcement Learning for Intelligent Congestion Control
3. AI-Based Intelligent Load Balancing in Software-Defined Networks Using Deep Reinforcement Learning
4. Explainable AI-Based Dynamic Routing Recommendation System for Intelligent and Secure Computer Networks
5. AI-Based Hybrid OSPF–BGP Optimization Using Reinforcement Learning for Adaptive Network Traffic Management
6. Deep Reinforcement Learning-Based Adaptive Congestion-Free Routing for QoS-Aware Computer Networks
7. LSTM-Based Internet Traffic Forecasting for Intelligent Bandwidth Prediction in Cloud, ISP, and Data Center Networks
8. Graph Neural Network-Based Network Congestion Prediction for Proactive Traffic Management in Computer Networks
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18. Deep Learning-Based Intelligent WiFi Channel Selection for Congestion-Aware Wireless Network Optimization
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22. AI-Based Video Quality of Service Prediction Using LSTM and Transformer for Intelligent Multimedia Networks
23. Deep Learning-Based Dynamic Bandwidth Allocation for Intelligent Network Resource Management
24. AI-Based Intelligent IPv6 Address Management for Automated Address Allocation and Scalable Network Administration
25. LLM-Based Intelligent Network Troubleshooting Assistant for Automated Log Analysis and Root Cause Diagnosis
26. LLM-Based Intelligent Cisco Packet Tracer Lab Generator for Automated Networking Education
27. LLM-Based AI Network Configuration Generator for Automated Cisco Network Configuration

Field	Details
Title 1	**AI-Based Intelligent Network Design Assistant for Automated Enterprise Network Planning**
Problem Statement	Designing an enterprise network requires network administrators to manually analyze business requirements and configure **VLANs, IP addressing, routing protocols, firewall policies, and QoS rules**. This process is time-consuming, requires expert knowledge, and is prone to configuration errors, leading to inefficient network performance and security vulnerabilities. Existing network design tools primarily assist with documentation and visualization but do not automatically generate optimized network architectures. This research proposes an **AI-Based Network Design Assistant** that accepts high-level organizational requirements as input and automatically generates a complete, optimized network design, including VLAN segmentation, IP addressing plans, routing strategies, firewall rules, and QoS policies.
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AI Model	**Large Language Models (LLMs) + Machine Learning + Knowledge Graphs** • GPT-based LLM (Network Design Reasoning) • Retrieval-Augmented Generation (RAG) • Graph Neural Networks (Optional for topology optimization) • Reinforcement Learning (Optional for design optimization)
Possible Inputs	• Number of departments • Number of employees • Office locations and branch topology • Floor plans/building layout • Network devices available • Number of servers • Security requirements • Internet bandwidth requirements • Voice, video, and data traffic requirements • QoS priorities • Wireless coverage requirements • Budget constraints • Scalability requirements • Regulatory/compliance requirements
Novel Contributions	• Develop an **AI-powered network design assistant** that converts business requirements into complete network architectures. • Automatically generate **VLAN design**, **IPv4/IPv6 addressing plans**, **routing protocol recommendations**, **firewall policies**, and **QoS configurations**. • Optimize network design based on scalability, security, performance, and cost constraints. • Generate configuration templates for major vendors (Cisco, Juniper, MikroTik, etc.). • Compare AI-generated designs with manually created enterprise network designs in terms of efficiency, cost, and performance.
Dataset	• Cisco Validated Design (CVD) documents • Enterprise network topology datasets (Internet Topology Zoo) • SNDlib Network Dataset • Public enterprise network configuration repositories • Mininet and GNS3 simulated enterprise network scenarios • Custom dataset created from enterprise case studies and network design best practices.
Research Gap	Existing network planning tools focus primarily on **visualization, documentation, or configuration management**, requiring network engineers to make all design decisions manually. Current AI applications in networking mainly optimize individual tasks such as routing or traffic engineering but rarely provide **end-to-end automated network design**. There is limited research on integrating **LLMs, networking knowledge bases, and optimization algorithms** to design complete enterprise networks from natural language requirements automatically.
Novelty	Proposes an **AI-driven enterprise network architect** that automatically transforms business requirements into deployable network designs, including **VLAN architecture, IP addressing plans, routing configurations, firewall policies, and QoS strategies**. Unlike conventional design tools, the proposed system combines **Large Language Models, networking knowledge, and AI-based optimization** to provide intelligent, explainable, and deployment-ready network designs. Future extensions can include **digital twins, intent-based networking, multi-cloud integration, zero-trust security design, and automatic configuration generation**.

Field	Details
Title 2	AI-Based TCP Congestion Window Prediction Using Deep Learning and Reinforcement Learning for Intelligent Congestion Control
Problem Statement	Traditional TCP congestion control algorithms such as Additive Increase Multiplicative Decrease (AIMD) adjust the congestion window (CWND) using predefined heuristic rules. While effective in conventional networks, these approaches struggle to adapt to dynamic network conditions, resulting in suboptimal throughput, increased latency, packet loss, and inefficient bandwidth utilization. This research proposes an AI-based TCP congestion window prediction framework that learns the optimal congestion window size from real-time network conditions, enabling adaptive congestion control that maximizes throughput while minimizing congestion, delay, and packet loss.
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AI Model	Deep Learning and Reinforcement Learning • Long Short-Term Memory (LSTM) • Transformer • Deep Reinforcement Learning (DQN/PPO/A2C)
Possible Inputs	• Current congestion window (CWND) • Round Trip Time (RTT) • Smoothed RTT (SRTT) • RTT variation (RTTVAR) • Packet loss rate • ACK arrival rate • Retransmission count • Throughput • Available bandwidth • Queue length • Link utilization • Network delay • Jitter • Historical TCP flow statistics
Novel Contributions	• Develop an AI-driven congestion window prediction model that replaces the fixed AIMD mechanism with intelligent, data-driven decision-making. • Compare LSTM, Transformer, and Reinforcement Learning approaches for adaptive congestion window prediction. • Design an RL reward function that jointly maximizes throughput while minimizing latency, packet loss, and retransmissions. • Evaluate performance under varying traffic loads, network congestion levels, and link failures. • Benchmark the proposed approach against conventional TCP variants such as TCP Reno, TCP NewReno, TCP Cubic, and TCP BBR.
Dataset	• MAWI Working Group Traffic Archive • CAIDA Internet Traffic Dataset • Internet2 Traffic Dataset • Pantheon TCP Benchmark Dataset • NS-3 generated TCP simulation datasets • Mininet or OMNeT++ generated datasets with multiple congestion scenarios and TCP variants.
Research Gap	Existing TCP congestion control mechanisms rely on handcrafted congestion control rules that cannot fully adapt to rapidly changing network environments. Although several AI-based TCP optimization methods have been proposed, most focus on a single learning technique or limited network conditions. There is limited research comparing LSTM, Transformer, and Reinforcement Learning for optimal congestion window prediction, while jointly optimizing throughput, latency, packet loss, and QoS across heterogeneous networks.
Novelty	Introduces an AI-based adaptive TCP congestion control framework that predicts the optimal congestion window (CWND) using LSTM, Transformer, and Reinforcement Learning instead of the traditional AIMD algorithm. The framework enables real-time, self-learning congestion control, improving throughput, reducing latency and packet loss, and enhancing QoS in dynamic network environments. Comparative evaluation of multiple AI paradigms further strengthens the contribution.

Field	Details
Title 3	**AI-Based Intelligent Load Balancing in Software-Defined Networks Using Deep Reinforcement Learning**
Problem Statement	Traditional load balancing techniques in Software-Defined Networking (SDN), such as **Round Robin**, **Least Connections**, and **Weighted Round Robin**, distribute traffic using predefined policies without considering real-time network conditions or server status. These approaches may lead to server overload, inefficient resource utilization, increased response time, and degraded Quality of Service (QoS) under dynamic workloads. This research proposes a **Deep Reinforcement Learning (DRL)-based intelligent load balancing framework** that continuously learns network and server conditions to dynamically select the **best server** for each incoming request, thereby optimizing resource utilization, minimizing response time, and improving overall network performance.
AI Model	**Deep Reinforcement Learning (DRL)** • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Soft Actor-Critic (SAC) (Optional Extension)
Possible Inputs	• Server CPU utilization • Memory utilization • Network bandwidth • Active connection count • Queue length • Response time • End-to-end latency • Packet loss rate • Throughput • Link utilization • Flow statistics from SDN controller • Server availability • Traffic demand • QoS requirements
Novel Contributions	• Develop an **AI-driven SDN load balancing framework** that replaces static algorithms such as Round Robin with adaptive decision making. • Design a **Deep Reinforcement Learning agent** capable of selecting the optimal server in real time. • Create a reward function that jointly optimizes throughput, latency, response time, server utilization, and load distribution. • Evaluate performance under varying workloads, traffic bursts, and server failures. • Compare the proposed framework with traditional load balancing algorithms and existing AI-based approaches.
Dataset	• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Internet Topology Zoo Dataset • SNDlib Network Dataset • Custom datasets generated using **Mininet, Ryu Controller, OpenFlow, Open vSwitch (OVS), and NS-3** with multiple servers and dynamic traffic scenarios.
Research Gap	Existing SDN load balancing methods primarily rely on **static or rule-based algorithms** that cannot adapt to rapidly changing network traffic and server conditions. Current AI-based solutions often optimize a single performance metric or operate in simulated environments with limited scalability. There is limited research on applying **Deep Reinforcement Learning** for **real-time, multi-objective server selection** that simultaneously optimizes response time, throughput, server utilization, QoS, and energy efficiency in SDN environments.
Novelty	Introduces a **Deep Reinforcement Learning-based intelligent load balancer** that continuously learns network and server conditions to dynamically select the optimal server instead of relying on static algorithms such as Round Robin. The framework enables **adaptive load balancing**, **real-time traffic engineering**, **QoS-aware resource allocation**, and **autonomous network optimization**. Additional novelty can be achieved by incorporating **multi-agent reinforcement learning**, **digital twins**, **energy-aware server selection**, or **Explainable AI** for transparent decision-making.

Field	Details
Title 4	**Explainable AI-Based Dynamic Routing Recommendation System for Intelligent and Secure Computer Networks**
Problem Statement	Traditional routing protocols such as **OSPF** and **RIP** select network paths using predefined static metrics (e.g., hop count or link cost), making them unsuitable for dynamic network environments. They cannot simultaneously consider multiple Quality of Service (QoS) and security parameters such as **delay, congestion, packet loss, energy consumption, and security risk**. Moreover, these protocols do not explain their routing decisions, making network management and troubleshooting difficult. This research aims to develop an **Explainable AI (XAI)-based routing recommendation system** that intelligently predicts the optimal routing path and provides human-understandable explanations for each routing decision.
AI Model	**Explainable AI (XAI)** using **SHAP (SHapley Additive Explanations)** and **LIME (Local Interpretable Model-Agnostic Explanations)** to explain route recommendations.
Possible Inputs	• Network delay (Latency) • Available bandwidth • Congestion level • Packet loss rate • Energy consumption of intermediate nodes • Security risk score • Link utilization • Queue length • Hop count • Historical traffic statistics • Link reliability • Network topology information
Novel Contributions	• Develop a **multi-criteria AI-based routing recommendation engine** considering both QoS and security metrics. • Integrate **Explainable AI** to justify every routing decision with feature importance analysis. • Design a **dynamic routing recommendation framework** adaptable to changing network conditions. • Compare the proposed system with traditional routing protocols (OSPF/RIP) and existing AI-based routing methods. • Improve network performance while maintaining transparency and trust in AI-driven routing decisions.
Dataset	• **Internet Topology Zoo Dataset • SNDlib Network Dataset • CAIDA Internet Traffic Dataset • NS-3 or OMNeT++ simulated routing datasets** • Custom dataset generated using **Mininet**, **GNS3**, or **Cisco Packet Tracer** simulations under different traffic and attack scenarios.
Research Gap	Existing AI-based routing approaches primarily focus on optimizing network performance but behave as **black-box models** and do not explain their routing recommendations. Traditional routing protocols rely on fixed metrics and fail to adapt to dynamic network conditions. There is limited research combining **dynamic routing optimization, security awareness, and Explainable AI** into a unified intelligent routing framework.
Novelty	Introduces one of the first **Explainable AI-driven routing recommendation systems** that simultaneously optimizes **delay, congestion, packet loss, energy consumption, and security risk**, while generating transparent, interpretable explanations for each routing decision using SHAP/LIME. This bridges the gap between intelligent routing and trustworthy AI.

Field	Details
Title 5	**AI-Based Hybrid OSPF–BGP Optimization Using Reinforcement Learning for Adaptive Network Traffic Management**
Problem Statement	Enterprise and Internet-scale networks commonly use **OSPF** for intra-domain routing and **BGP** for inter-domain routing. Routing policies are typically configured manually by network administrators based on static rules, making them difficult to adapt to dynamic traffic conditions, link failures, congestion, and changing Quality of Service (QoS) requirements. This research aims to develop a **Reinforcement Learning (RL)-based intelligent routing controller** that dynamically determines whether traffic should be forwarded using **OSPF, BGP, or an alternate routing path** based on real-time network conditions, thereby improving network performance, resilience, and resource utilization without manual intervention.
AI Model	**Reinforcement Learning (RL)** • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Deep Deterministic Policy Gradient (DDPG) (for continuous optimization)
Possible Inputs	• Network traffic load • Link utilization • End-to-end delay • Packet loss rate • Available bandwidth • Queue length • Network congestion level • Link failures • OSPF link-state information • BGP route advertisements and path attributes (AS Path, Local Preference, MED) • Historical routing performance • QoS requirements of applications
Novel Contributions	• Design an **AI-driven hybrid routing controller** that intelligently selects between OSPF, BGP, and alternate paths. • Replace static routing policies with **adaptive Reinforcement Learning-based decision making**. • Develop a reward function that jointly optimizes latency, throughput, packet delivery, congestion, and routing stability. • Enable self-learning from changing network conditions without manual policy updates. • Demonstrate improved network resilience during congestion and link failures compared to traditional routing approaches.
Dataset	• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Internet Topology Zoo Dataset • SNDlib Network Dataset • RIPE Routing Information Service (RIS) BGP data • RouteViews BGP Dataset • Custom datasets generated using **Mininet, GNS3, NS-3, OMNeT++, or Cisco Packet Tracer** under varying traffic scenarios.
Research Gap	Existing hybrid OSPF–BGP networks rely on **manually configured routing policies**, which cannot respond effectively to rapidly changing network conditions. Current AI-based routing solutions generally optimize a single protocol or a limited set of metrics and rarely address **dynamic protocol selection** between OSPF and BGP using Reinforcement Learning. There is limited research on self-learning controllers capable of making real-time routing decisions across multiple routing protocols.
Novelty	Proposes a **Reinforcement Learning-based adaptive hybrid routing framework** that autonomously selects **OSPF, BGP, or alternate paths** according to real-time network conditions. Unlike traditional static routing policies, the proposed system continuously learns optimal routing strategies, improving adaptability, QoS, and network resilience while reducing administrative overhead.

Field	Details
Title 6	**Deep Reinforcement Learning-Based Adaptive Congestion-Free Routing for QoS-Aware Computer Networks**
Problem Statement	Traditional routing algorithms primarily select the **shortest path** based on static metrics such as hop count or link cost. However, the shortest path is not always the optimal path under dynamic traffic conditions, often resulting in **network congestion, increased delay, packet losses, and reduced throughput**. Existing routing protocols cannot learn from changing network environments and adapt routing decisions in real time. This research proposes a **Deep Reinforcement Learning (DRL)-based adaptive routing framework** that continuously learns optimal routing policies to maximize throughput while minimizing end-to-end delay, packet drops, and congestion, thereby improving overall Quality of Service (QoS).
AI Model	**Deep Reinforcement Learning (DRL)** • **Deep Q-Network (DQN)** (Primary Model) • Optional comparison with Double DQN (DDQN) and Dueling DQN
Possible Inputs	• Network topology • Link utilization • Queue length • Available bandwidth • End-to-end delay • Packet loss rate • Congestion level • Hop count • Throughput • TCP congestion window (CWND) • Round Trip Time (RTT) • Link reliability • Historical traffic patterns
Novel Contributions	• Develop a **Deep Q-Network-based intelligent routing agent** that learns optimal routing decisions through continuous interaction with the network. • Replace shortest-path routing with **adaptive QoS-aware routing** optimized for real-time traffic conditions. • Design a reward function that simultaneously maximizes throughput while minimizing delay, congestion, and packet drops. • Evaluate performance under varying traffic loads, congestion scenarios, and link failures. • Compare the proposed approach with conventional routing protocols (OSPF, RIP) and shortest-path algorithms.
Dataset	• CAIDA Internet Traffic Dataset • MAWI Traffic Archive • Internet Topology Zoo Dataset • SNDlib Network Dataset • NS-3 simulation datasets • OMNeT++ simulation datasets • Custom datasets generated using **Mininet, GNS3, Cisco Packet Tracer, or NS-3** with different congestion and traffic conditions.
Research Gap	Existing routing protocols optimize only static metrics and cannot adapt to changing traffic conditions. Many AI-based routing solutions focus on a single QoS parameter or require extensive manual tuning. There is limited research on applying **Deep Reinforcement Learning** to simultaneously optimize **throughput, delay, packet loss, and congestion** while integrating routing, congestion control, QoS, and TCP behavior into a unified intelligent routing framework.
Novelty	Introduces a **Deep Reinforcement Learning-based adaptive routing framework** that learns congestion-aware routing policies instead of relying on shortest-path algorithms. The proposed system jointly optimizes **maximum throughput, minimum delay, minimum packet loss, and congestion avoidance**, enabling self-learning and real-time adaptation in dynamic network environments.

Field	Details
Title 7	**LSTM-Based Internet Traffic Forecasting for Intelligent Bandwidth Prediction in Cloud, ISP, and Data Center Networks**
Problem Statement	Modern communication networks such as **Cloud Computing environments, Internet Service Providers (ISPs), and Data Centers** experience highly dynamic and unpredictable traffic patterns. Traditional bandwidth allocation techniques are primarily reactive, resulting in network congestion, poor resource utilization, and degraded Quality of Service (QoS). Accurate forecasting of future network traffic can enable proactive bandwidth allocation and efficient network management. This research proposes an **AI-based Internet Traffic Forecasting framework** using deep learning models to predict future bandwidth demand, enabling intelligent resource provisioning and congestion avoidance.
AI Model	**Deep Learning Models • Long Short-Term Memory (LSTM)** (Primary Model) • **Gated Recurrent Unit (GRU) • Transformer-based Time Series Model**
Possible Inputs	• Historical bandwidth utilization • Network throughput • Packet arrival rate • Packet size distribution • Link utilization • Network latency • Time of day/week • Number of active users • CPU and memory utilization of network devices • Traffic type (HTTP, Video Streaming, VoIP, FTP, etc.) • QoS statistics • Network event logs
Novel Contributions	• Develop an **AI-based bandwidth forecasting framework** using sequential deep learning models. • Compare the performance of **LSTM, GRU, and Transformer** models for short-term and long-term traffic prediction. • Design a proactive bandwidth allocation strategy based on predicted traffic demand. • Improve QoS by reducing congestion and optimizing network resource utilization. • Analyze model interpretability and computational efficiency for deployment in real-world networking environments.
Dataset	• MAWI Working Group Traffic Archive • CAIDA Internet Traffic Dataset • GEANT Network Traffic Dataset • Abilene Network Traffic Dataset • Internet2 Traffic Dataset • UGR'16 Network Traffic Dataset • Custom datasets collected from NS-3, OMNeT++, Mininet, or SDN testbeds.
Research Gap	Existing traffic prediction models often focus on a single deep learning architecture and are evaluated on limited network scenarios. Many approaches fail to capture long-term temporal dependencies and changing traffic patterns while providing limited comparison across modern sequence models. There is a need for a comprehensive framework that evaluates **LSTM, GRU, and Transformer** models for accurate bandwidth prediction across diverse networking environments such as **Cloud, ISP, and Data Centers**, with an emphasis on proactive resource management.
Novelty	Proposes a **deep learning-based intelligent traffic forecasting framework** that systematically compares **LSTM, GRU, and Transformer** architectures for accurate bandwidth prediction. The framework supports **proactive bandwidth allocation, congestion prevention, and intelligent resource management** across Cloud, ISP, and Data Center networks.

Field	Details
Title 8	**Graph Neural Network-Based Network Congestion Prediction for Proactive Traffic Management in Computer Networks**
Problem Statement	Modern communication networks are naturally represented as **graphs**, where **routers/switches are nodes** and **communication links are edges**. Traditional congestion detection techniques identify congestion only after it has occurred, leading to increased latency, packet loss, reduced throughput, and degraded Quality of Service (QoS). Existing machine learning approaches often ignore the underlying graph topology of the network, limiting their prediction accuracy. This research proposes a **Graph Neural Network (GNN)-based congestion prediction framework** that learns both the structural relationships and traffic dynamics of the network to **predict future congestion before it occurs**, enabling proactive traffic engineering, intelligent routing, and efficient resource allocation.
AI Model	**Graph Neural Networks (GNNs)** • Graph Convolutional Network (GCN) • Graph Attention Network (GAT) • GraphSAGE • Spatio-Temporal Graph Neural Network (ST-GNN) (Primary Recommendation) • Temporal Graph Network (TGN) (Optional Comparison)
Possible Inputs	• Network topology (Graph structure) • Router and switch information (Nodes) • Communication links (Edges) • Link utilization • Queue length • Available bandwidth • Packet arrival rate • Throughput • End-to-end delay • Packet loss rate • CPU and memory utilization of routers • Historical congestion patterns • Traffic flow statistics • Link failures • QoS metrics
Novel Contributions	• Develop a **Graph Neural Network-based congestion prediction model** that explicitly captures network topology. • Predict congestion **before it occurs**, enabling proactive congestion avoidance instead of reactive congestion control. • Compare multiple GNN architectures (GCN, GAT, GraphSAGE, ST-GNN) for congestion prediction accuracy. • Integrate predicted congestion information into intelligent routing or traffic engineering frameworks. • Improve network throughput, reduce packet loss, and minimize end-to-end delay through early congestion prediction.
Dataset	• Internet Topology Zoo Dataset • SNDlib Network Dataset • CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • GEANT Traffic Dataset • Abilene Network Dataset • Custom datasets generated using **NS-3, OMNeT++, Mininet, GNS3, or SDN testbeds** with different congestion scenarios.
Research Gap	Existing congestion prediction techniques primarily rely on traditional machine learning or time-series models that do not exploit the **graph structure** of communication networks. Many approaches treat network devices independently, ignoring spatial dependencies between routers and links. Furthermore, limited research integrates **spatio-temporal graph learning** for **early congestion prediction** in large-scale communication networks, leaving significant scope for improving prediction accuracy and proactive traffic management.

Field	Details
Title 9	Federated Learning-Based Privacy-Preserving Internet Traffic Prediction for Multi-ISP Networks
Problem Statement	Internet Service Providers (ISPs) generate massive volumes of network traffic data that can be used to improve traffic prediction, bandwidth allocation, and Quality of Service (QoS). However, sharing user traffic data across ISPs raises significant privacy, security, and regulatory concerns. Existing centralized machine learning approaches require aggregating data from multiple organizations, increasing the risk of sensitive information leakage. This research proposes a Federated Learning (FL)-based traffic prediction framework that enables multiple ISPs to collaboratively train a common AI model without sharing raw user data, thereby preserving privacy while improving traffic prediction accuracy and network resource management.
AI Model	Federated Learning (FL) • Federated Averaging (FedAvg) • FedProx • FedNova • LSTM/GRU-based local models • Transformer-based local models (optional)
Possible Inputs	• Historical network traffic data • Bandwidth utilization • Packet arrival rate • Throughput • Link utilization • Latency • Packet loss rate • Time-of-day and day-of-week information • Active user count • Application traffic categories (VoIP, Video, Web, Gaming, etc.) • QoS metrics • ISP-specific network statistics
Novel Contributions	• Develop a privacy-preserving federated traffic prediction framework for multiple ISPs. • Eliminate the need to exchange raw traffic data while maintaining high prediction accuracy. • Compare centralized learning, local learning, and federated learning approaches. • Evaluate communication overhead, model convergence, scalability, and privacy preservation. • Investigate the robustness of the federated model under heterogeneous ISP traffic distributions (non-IID data).
Dataset	• MAWI Working Group Traffic Archive • CAIDA Internet Traffic Dataset • GEANT Traffic Dataset • Abilene Network Traffic Dataset • Internet2 Traffic Dataset • UGR'16 Network Traffic Dataset • Custom federated datasets created by partitioning real traffic datasets to simulate multiple ISPs with heterogeneous traffic patterns.
Research Gap	Most existing traffic prediction models rely on centralized machine learning, requiring all traffic data to be collected at a central server. Such approaches raise privacy concerns and are often impractical due to regulatory restrictions and commercial competition among ISPs. Current federated learning research has focused mainly on mobile devices and healthcare, with relatively limited work on privacy-preserving traffic prediction across multiple ISPs, particularly under non-IID traffic distributions, communication constraints, and large-scale network environments.
Novelty	Introduces a Federated Learning-based collaborative traffic prediction framework that enables multiple ISPs to jointly train a high-performance prediction model without sharing raw user traffic data. The framework combines privacy preservation, distributed AI, and intelligent traffic forecasting, making it suitable for next-generation secure networking environments. Extensions such as secure aggregation, differential privacy, or personalized federated learning can further enhance the novelty and practical impact.

Field	Details
Title 10	**Deep Learning-Based Adaptive Retransmission Timeout Estimation for Intelligent TCP Communication**
Problem Statement	Traditional TCP retransmission timeout (RTO) estimation algorithms, such as **Jacobson's RTT Estimator** and **Karn's Algorithm**, compute the retransmission timeout using predefined statistical formulas based on historical Round Trip Time (RTT) measurements. Although effective in conventional networks, these approaches struggle to accurately estimate RTO under highly dynamic conditions such as wireless networks, mobile environments, cloud infrastructures, and high-speed networks. Inaccurate RTO estimation leads to unnecessary retransmissions, increased latency, reduced throughput, and poor Quality of Service (QoS). This research proposes a **Deep Learning-based adaptive retransmission timer** that predicts **RTT** and **optimal Retransmission Timeout (RTO)** using historical network behavior and real-time traffic characteristics.
AI Model	**Deep Learning Models** • Long Short-Term Memory (LSTM) • Gated Recurrent Unit (GRU) • Transformer-based Time Series Model
Possible Inputs	• Historical RTT measurements • Smoothed RTT (SRTT) • RTT variation (RTTVAR) • Current Retransmission Timeout (RTO) • Packet loss rate • ACK arrival intervals • Network latency • Jitter • Queue length • Link utilization • Throughput • TCP congestion window (CWND) • Network congestion level
Novel Contributions	• Develop an **AI-driven adaptive RTO estimator** replacing Jacobson and Karn algorithms. • Predict RTT and RTO using sequential deep learning models. • Compare LSTM, GRU, and Transformer architectures for RTO estimation accuracy. • Reduce unnecessary retransmissions and improve TCP throughput. • Evaluate performance under varying network delays, congestion, and wireless conditions.
Dataset	• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Pantheon TCP Benchmark Dataset • Internet2 Traffic Dataset • NS-3 generated TCP datasets • Mininet/OMNeT++ simulation datasets under different RTT and congestion scenarios.
Research Gap	Existing retransmission timeout estimation techniques rely on statistical heuristics that cannot effectively capture complex temporal network behaviors. Most AI-based studies focus on congestion control rather than **adaptive RTT and RTO prediction**. There is limited research comparing modern deep learning models for intelligent retransmission timer estimation across diverse network environments.
Novelty	Introduces a **deep learning-based adaptive retransmission timer** that replaces conventional Jacobson and Karn algorithms with **LSTM, GRU, and Transformer** models for intelligent RTT and RTO prediction. The framework enables more accurate timeout estimation, reducing retransmissions and improving TCP performance in dynamic communication networks.

Field	Details
Title 11	**AI-Based Intelligent Load Balancing in Software-Defined Networks Using Deep Reinforcement Learning**
Problem Statement	Traditional load balancing techniques in Software-Defined Networking (SDN), such as **Round Robin**, **Least Connections**, and **Weighted Round Robin**, distribute traffic using predefined policies without considering real-time network conditions or server status. These approaches may lead to server overload, inefficient resource utilization, increased response time, and degraded Quality of Service (QoS) under dynamic workloads. This research proposes a **Deep Reinforcement Learning (DRL)-based intelligent load balancing framework** that continuously learns network and server conditions to dynamically select the **best server** for each incoming request, thereby optimizing resource utilization, minimizing response time, and improving overall network performance.
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AI Model	**Deep Reinforcement Learning (DRL)** • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Soft Actor-Critic (SAC) (Optional Extension)
Possible Inputs	• Server CPU utilization • Memory utilization • Network bandwidth • Active connection count • Queue length • Response time • End-to-end latency • Packet loss rate • Throughput • Link utilization • Flow statistics from SDN controller • Server availability • Traffic demand • QoS requirements
Novel Contributions	• Develop an **AI-driven SDN load balancing framework** that replaces static algorithms such as Round Robin with adaptive decision making. • Design a **Deep Reinforcement Learning agent** capable of selecting the optimal server in real time. • Create a reward function that jointly optimizes throughput, latency, response time, server utilization, and load distribution. • Evaluate performance under varying workloads, traffic bursts, and server failures. • Compare the proposed framework with traditional load balancing algorithms and existing AI-based approaches.
Dataset	• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Internet Topology Zoo Dataset • SNDlib Network Dataset • Custom datasets generated using **Mininet, Ryu Controller, OpenFlow, Open vSwitch (OVS), and NS-3** with multiple servers and dynamic traffic scenarios.
Research Gap	Existing SDN load balancing methods primarily rely on **static or rule-based algorithms** that cannot adapt to rapidly changing network traffic and server conditions. Current AI-based solutions often optimize a single performance metric or operate in simulated environments with limited scalability. There is limited research on applying **Deep Reinforcement Learning** for **real-time, multi-objective server selection** that simultaneously optimizes response time, throughput, server utilization, QoS, and energy efficiency in SDN environments.
Novelty	Introduces a **Deep Reinforcement Learning-based intelligent load balancer** that continuously learns network and server conditions to dynamically select the optimal server instead of relying on static algorithms such as Round Robin. The framework enables **adaptive load balancing**, **real-time traffic engineering**, **QoS-aware resource allocation**, and **autonomous network optimization**. Additional novelty can be achieved by incorporating **multi-agent reinforcement learning**, **digital twins**, **energy-aware server selection**, or **Explainable AI** for transparent decision making.

Field	Details
Title 12	Reinforcement Learning-Based Intelligent SDN Controller for Autonomous Traffic Management and Flow Table Optimization
Problem Statement	Software-Defined Networking (SDN) separates the control plane from the data plane, enabling centralized network management through an SDN controller. However, existing SDN controllers typically rely on static routing policies and predefined traffic engineering rules, making them less effective in handling dynamic traffic patterns, congestion, and link failures. Manual flow table configuration also increases management complexity and reduces network adaptability. This research proposes a Reinforcement Learning (RL)-based intelligent SDN controller that continuously learns network traffic conditions and automatically reconfigures OpenFlow flow tables to optimize routing, load balancing, congestion control, and Quality of Service (QoS) without human intervention.
AI Model	Deep Reinforcement Learning (DRL) • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Soft Actor-Critic (SAC) (Optional Extension)
Possible Inputs	• Network topology • Flow table statistics • Link utilization • Available bandwidth • Network throughput • End-to-end delay • Packet loss rate • Queue length • Switch CPU utilization • OpenFlow statistics • Active flow count • Link failures • Traffic demand • QoS requirements
Novel Contributions	• Develop a self-learning SDN controller capable of autonomous flow table management. • Replace static OpenFlow rule configuration with adaptive Reinforcement Learning-based optimization. • Design a reward function that jointly optimizes throughput, latency, packet loss, load balancing, and controller overhead. • Enable real-time traffic engineering through intelligent flow rerouting and congestion avoidance. • Compare the proposed RL controller with traditional SDN routing algorithms and heuristic traffic engineering methods.
Dataset	• Mininet-generated traffic datasets • Internet Topology Zoo Dataset • CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • SNDlib Network Dataset • Custom datasets generated using Mininet, Ryu Controller, OpenFlow, Open vSwitch (OVS), and NS-3 under different traffic loads, congestion scenarios, and link failures.
Research Gap	Most SDN controllers rely on rule-based traffic engineering and static optimization techniques that cannot effectively adapt to dynamic network environments. Existing AI-based SDN solutions often focus on isolated tasks such as routing or load balancing without providing a comprehensive autonomous controller capable of continuous learning and automatic flow table reconfiguration. Furthermore, limited research evaluates advanced Deep Reinforcement Learning algorithms for real-time SDN traffic management under diverse network conditions.
Novelty	Introduces a Deep Reinforcement Learning-based autonomous SDN controller that continuously learns network behavior and automatically updates OpenFlow flow tables without predefined policies. By integrating traffic prediction, intelligent routing, congestion avoidance, and adaptive traffic engineering, the proposed framework enables self-optimizing SDN networks. Extensions such as multi-agent reinforcement learning, digital twins, or federated reinforcement learning can further enhance novelty and scalability.

Field	Details
Title 13	**Graph Neural Network-Based Traffic Engineering for Intelligent Software-Defined Networks**
Problem Statement	Software-Defined Networking (SDN) enables centralized network control, allowing traffic engineering decisions to be implemented dynamically through the SDN controller. However, existing traffic engineering approaches often rely on **shortest-path algorithms, heuristic optimization, or predefined routing policies**, which fail to fully exploit the complex topological relationships of communication networks. These methods struggle under dynamic traffic conditions, leading to congestion, poor load distribution, increased latency, and reduced Quality of Service (QoS). This research proposes a **Graph Neural Network (GNN)-based traffic engineering framework** that learns the **network topology and traffic dynamics** to intelligently optimize routing decisions, maximize network utilization, and proactively avoid congestion.
AI Model	**Graph Neural Networks (GNNs)** • Graph Convolutional Network (GCN) • Graph Attention Network (GAT) • GraphSAGE • Spatio-Temporal Graph Neural Network (ST-GNN) (Primary Recommendation) • Graph Transformer Network (GTN) (Optional Extension)
Possible Inputs	• SDN network topology (Graph representation) • Switches and routers (Nodes) • Communication links (Edges) • Flow statistics • Link utilization • Available bandwidth • End-to-end delay • Packet loss rate • Queue length • Throughput • Network congestion level • OpenFlow statistics • Historical traffic matrices • QoS requirements
Novel Contributions	• Develop a **Graph Neural Network-based traffic engineering framework** for SDN. • Learn both **network topology** and **traffic characteristics** simultaneously for intelligent routing. • Optimize routing decisions to improve throughput, reduce latency, and avoid congestion. • Compare multiple GNN architectures (GCN, GAT, GraphSAGE, ST-GNN) for SDN traffic engineering. • Integrate the trained GNN with an SDN controller (e.g., Ryu/ONOS) for real-time route optimization and OpenFlow rule updates.
Dataset	• Internet Topology Zoo Dataset • SNDlib Network Dataset • CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • GEANT Traffic Dataset • Mininet-generated SDN datasets • Custom datasets collected using **Mininet, Ryu Controller, OpenFlow, Open vSwitch (OVS), NS-3, or OMNeT++** under varying traffic loads and congestion scenarios.
Research Gap	Existing SDN traffic engineering methods primarily use **rule-based optimization, shortest-path routing, or conventional machine learning**, which do not effectively capture the graph structure of communication networks. Although Graph Neural Networks have demonstrated success in graph-structured domains, their application to **real-time SDN traffic engineering** remains limited. There is a need for intelligent routing frameworks that jointly learn **network topology, traffic dynamics, and routing optimization** for autonomous SDN management.
Novelty	Proposes a **Graph Neural Network-driven SDN traffic engineering framework** that models communication networks as graphs and learns **topological relationships** together with **traffic behavior** to optimize routing decisions. Unlike traditional traffic engineering methods, the proposed approach performs **topology-aware intelligent routing**, enabling adaptive congestion avoidance, load balancing, and QoS optimization. Further enhancements using **Graph Transformers**, **Multi-Agent Reinforcement Learning**, or **Digital Twin-assisted SDN optimization** can significantly increase the novelty and research impact.

Field	Details
Title 14	**Transformer-Based Network Intrusion Detection System for Intelligent Multi-Class Cyber Attack Classification**
Problem Statement	Traditional Network Intrusion Detection Systems (NIDS) primarily rely on **signature-based detection** or conventional machine learning techniques, which often struggle to detect sophisticated, evolving, and zero-day attacks. Existing deep learning models such as CNNs and LSTMs capture either spatial or temporal patterns but may fail to model long-range dependencies within network traffic. This research proposes a **Transformer-based Network Intrusion Detection System (NIDS)** that leverages the self-attention mechanism to learn complex traffic patterns and accurately classify multiple cyber attacks, including **DoS, DDoS, Botnet, and Port Scan**, thereby improving detection accuracy, scalability, and real-time network security.
AI Model	**Transformer-Based Deep Learning** • Vision Transformer (ViT) (Optional for flow-image representations) • Temporal Transformer • TabTransformer • BERT-style Transformer Encoder (Primary Recommendation) • Transformer + Explainable AI (SHAP/LIME) (Optional Extension)
Possible Inputs	• Source IP address • Destination IP address • Source and destination ports • Protocol type • Flow duration • Packet size statistics • Packet inter-arrival time • Forward and backward packet counts • TCP flags • Flow bytes per second • Packet rate • Connection state • Network traffic features (CICFlowMeter features)
Novel Contributions	• Develop a **Transformer-based NIDS** for multi-class cyber attack detection. • Capture long-range dependencies in network traffic using the self-attention mechanism. • Compare Transformer performance with CNN, LSTM, GRU, and traditional machine learning models. • Detect multiple attack categories, including **DoS, DDoS, Botnet, Port Scan**, and optionally brute-force and infiltration attacks. • Improve detection accuracy while reducing false positives and enabling real-time deployment.
Dataset	**• CIC-IDS2017 • CSE-CIC-IDS2018 • UNSW-NB15 • TON_IoT Dataset • Bot-IoT Dataset • CIC-DDoS2019 • NF-UQ-NIDS Dataset** (Recommended) • **Custom datasets** generated using Mininet, NS-3, or SDN testbeds.
Research Gap	Most existing intrusion detection systems rely on **CNNs, LSTMs, or conventional machine learning**, which have limited capability in modeling long-range relationships among network traffic features. While Transformers have demonstrated exceptional performance in natural language processing and computer vision, their application to **network intrusion detection**, particularly for **multi-class attack classification and real-time deployment**, remains relatively limited. There is also a need to improve explainability and computational efficiency for practical cybersecurity applications.
Novelty	Proposes a **Transformer-based Network Intrusion Detection System** that exploits the **self-attention mechanism** to model complex traffic relationships for **multi-class cyber attack detection**. Unlike CNN- or LSTM-based IDS solutions, the proposed framework captures global dependencies in network traffic, improving detection accuracy for **DoS, DDoS, Botnet, Port Scan**, and other advanced attacks. Novelty can be further enhanced by integrating **Explainable AI (XAI)**, **Federated Learning**, **Graph Neural Networks**, or **Online Learning** for adaptive intrusion detection.

Field	Details
Title 15	**Explainable AI-Based Intelligent Firewall for Transparent Network Packet Filtering and Threat Detection**
Problem Statement	Traditional firewalls rely on **static rule-based filtering**, requiring continuous manual updates to detect emerging cyber threats. Although AI-based firewalls can automatically classify malicious traffic, most operate as **black-box models**, making it difficult for network administrators to understand why a packet was allowed or blocked. This lack of transparency reduces trust, complicates security audits, and limits adoption in critical infrastructures. This research proposes an **Explainable AI (XAI)-based Intelligent Firewall** that automatically predicts whether an incoming packet should be **allowed or blocked** while providing a clear, human-understandable explanation for every decision. The proposed framework enhances network security, transparency, and administrator confidence by combining intelligent threat detection with interpretable AI.
AI Model	**Explainable AI (XAI)** + **Machine Learning / Deep Learning** • XGBoost • Random Forest • LightGBM • Multi-Layer Perceptron (MLP) • Transformer (Optional Extension) • **SHAP (SHapley Additive Explanations)** • **LIME (Local Interpretable Model-Agnostic Explanations)**
Possible Inputs	• Source IP address • Destination IP address • Source and destination ports • Protocol type (TCP/UDP/ICMP) • Packet size • TCP flags (SYN, ACK, FIN, RST, etc.) • Flow duration • Packet rate • Connection state • Payload characteristics (optional) • Historical traffic behavior • Threat intelligence score • Blacklist/Whitelist information
Novel Contributions	• Develop an **AI-based intelligent firewall** capable of adaptive packet filtering. • Integrate **Explainable AI** to justify every allow/block decision using feature importance analysis. • Compare multiple ML/DL models for packet classification accuracy and interpretability. • Reduce false positives and false negatives while maintaining transparent security decisions. • Generate administrator-friendly explanations suitable for security auditing and compliance.
Dataset	• **CIC-IDS2017** • **CSE-CIC-IDS2018** • **UNSW-NB15** • **TON_IoT Dataset** • **Bot-IoT Dataset** • **CIC-DDoS2019** • **NF-UQ-NIDS Dataset** • Custom datasets generated using **Mininet**, **NS-3**, **OpenFlow/SDN testbeds**, or enterprise firewall logs.
Research Gap	Existing intelligent firewalls mainly focus on improving detection accuracy but provide **little or no explanation** for their decisions. Most AI-based packet filtering systems are black-box models, making them unsuitable for environments requiring transparency, regulatory compliance, and human verification. There is limited research integrating **Explainable AI** with **real-time firewall decision-making**, particularly for multi-class threat detection and administrator-friendly reasoning.
Novelty	Proposes an **Explainable AI-powered firewall** that not only classifies network packets as **allow** or **block**, but also generates **human-readable explanations** describing the key factors influencing each decision. Unlike conventional AI firewalls, the proposed framework combines **high detection accuracy**, **decision transparency**, and **security policy interpretability**, making it suitable for enterprise networks, cloud environments, and critical infrastructure. Future enhancements can integrate **Zero Trust Architecture**, **Federated Learning**, **Transformer-based packet classification**, or **Large Language Models (LLMs)** for natural-language security explanations.

Field	Details
Title 16	**AI-Based DNS Attack Detection System for Intelligent Detection of DNS Tunneling, Amplification, and Cache Poisoning Attacks**
Problem Statement	The **Domain Name System (DNS)** is one of the most critical Internet services and has become a major target for cyber attacks. Traditional DNS security mechanisms rely on **signature-based detection** and manually defined rules, making them ineffective against sophisticated and evolving attacks such as **DNS tunneling, DNS amplification, and DNS cache poisoning**. These attacks can be used for data exfiltration, Distributed Denial-of-Service (DDoS), malware communication, and phishing. This research proposes an **AI-based DNS Attack Detection System** that automatically analyzes DNS traffic to accurately detect and classify malicious DNS activities in real time, improving network security and reducing false alarms.
AI Model	**Machine Learning & Deep Learning** • XGBoost • Random Forest • LightGBM • Long Short-Term Memory (LSTM) • Transformer (Optional Extension) • Graph Neural Network (Optional for DNS relationship analysis)
Possible Inputs	• Source IP address • Destination IP address • DNS query name (domain) • Query type (A, AAAA, MX, TXT, CNAME, etc.) • DNS response code • Query frequency • Domain entropy • Query length • Response size • Time-to-Live (TTL) values • Inter-arrival time • Number of unique domains queried • NXDOMAIN response rate • Packet size • Flow statistics
Novel Contributions	• Develop an **AI-driven DNS attack detection framework** capable of identifying multiple DNS attack categories. • Detect **DNS tunneling, DNS amplification, and DNS cache poisoning** using intelligent traffic analysis. • Compare multiple ML and DL models for detection accuracy and computational efficiency. • Reduce false positives while enabling real-time DNS threat detection. • Extend the framework with Explainable AI (XAI) for transparent attack classification (optional).
Dataset	• **CIC-DNS-2021 Dataset** (Recommended) • **CIRA-CIC-DoHBrw-2020 Dataset** • **CIC-IDS2017** (DNS traffic subset) • **UNSW-NB15** • **TON_IoT Dataset** • **MAWI Traffic Archive** (DNS flows) • Custom DNS datasets generated using **BIND**, **Mininet**, **NS-3**, **Wireshark**, or enterprise DNS logs.
Research Gap	Existing DNS security solutions primarily depend on **signature-based detection** or handcrafted statistical features, limiting their ability to identify **unknown and evolving DNS attacks**. Many studies focus on a single attack type, such as DNS tunneling, without providing a unified framework for detecting multiple DNS threats. Furthermore, limited research integrates **deep learning**, **traffic behavior analysis**, and **real-time DNS attack classification** in a scalable and adaptive detection system.
Novelty	Proposes a **unified AI-based DNS security framework** capable of detecting **DNS tunneling, DNS amplification, and DNS cache poisoning** from network traffic using intelligent feature learning. Unlike conventional DNS security tools that rely on static signatures, the proposed framework adapts to evolving attack patterns and supports **real-time threat detection**. Novelty can be further enhanced by integrating **Transformer-based sequential analysis**, **Graph Neural Networks for DNS relationship modeling**, **Federated Learning**, or **Explainable AI** for transparent security decisions.

Field	Details
Title 17	**Deep Autoencoder-Based Zero-Day Attack Detection Using Unsupervised Anomaly Detection for Intelligent Network Security**
Problem Statement	Traditional Intrusion Detection Systems (IDS) rely heavily on **signature-based detection**, making them ineffective against **zero-day attacks** whose signatures are unknown. Supervised machine learning models also require labeled attack data, which is often unavailable for newly emerging threats. As cyber attacks continue to evolve, there is a growing need for intelligent systems capable of detecting previously unseen attacks without relying on predefined signatures. This research proposes a **Deep Autoencoder-based anomaly detection framework** that learns the normal behavior of network traffic and identifies deviations as potential **zero-day attacks**, enabling early detection of novel cyber threats while minimizing false alarms.
AI Model	**Deep Learning (Unsupervised Learning)** • Deep Autoencoder (Primary Model) • Variational Autoencoder (VAE) (Optional Comparison) • Denoising Autoencoder (DAE) • Sparse Autoencoder • Transformer Autoencoder (Optional Extension)
Possible Inputs	• Source IP address • Destination IP address • Source and destination ports • Protocol type • Flow duration • Packet size statistics • Packet arrival rate • Flow bytes/packets per second • TCP flags • Connection state • Network latency • Packet loss rate • CICFlowMeter network traffic features
Novel Contributions	• Develop a **signature-free zero-day attack detection framework** using deep autoencoders. • Learn normal network behavior through **unsupervised anomaly detection**, eliminating the need for labeled attack data. • Compare Deep Autoencoder, Variational Autoencoder, and Denoising Autoencoder models for anomaly detection performance. • Detect unknown cyber attacks with high detection accuracy and low false-positive rates. • Evaluate robustness under evolving attack scenarios and concept drift.
Dataset	**• CIC-IDS2017 • CSE-CIC-IDS2018 • UNSW-NB15 • TON_IoT Dataset • Bot-IoT Dataset • NF-UQ-NIDS Dataset • CIC-DDoS2019** • Custom datasets generated using **Mininet**, **NS-3**, **Wireshark**, or SDN testbeds with unseen attack scenarios.
Research Gap	Most existing IDS solutions rely on **signature databases** or **supervised learning**, limiting their ability to detect previously unseen attacks. Although autoencoders have been applied to anomaly detection, many studies evaluate them on limited datasets or fail to address **concept drift, evolving attack behavior, and real-time deployment**. There is a need for a scalable, adaptive, and unsupervised framework capable of accurately identifying **zero-day attacks** across diverse network environments.
Novelty	Proposes a **deep autoencoder-based, signature-free intrusion detection framework** that detects **zero-day attacks** by learning normal network behavior and identifying anomalies through reconstruction error. Unlike conventional IDS solutions, the proposed approach does not require prior knowledge of attack signatures or labeled attack samples. The novelty can be further enhanced by integrating **Explainable AI (XAI)**, **Federated Learning**, **Graph Neural Networks**, **Transformer Autoencoders**, or **Online Continual Learning** for adaptive and interpretable zero-day detection.

Field	Details
Title 18	**Deep Learning-Based Intelligent WiFi Channel Selection for Congestion-Aware Wireless Network Optimization**
Problem Statement	WiFi networks commonly operate using **fixed or manually selected channels**, which often results in channel interference, congestion, reduced throughput, increased latency, and poor Quality of Service (QoS), especially in dense environments such as offices, campuses, apartments, and public hotspots. Traditional channel selection techniques rely on static rules or periodic scanning and cannot adapt effectively to rapidly changing wireless conditions. This research proposes a **Deep Learning-based intelligent WiFi channel selection framework** that continuously analyzes wireless network conditions and predicts the **least congested channel** for optimal communication, thereby improving network performance and user experience.
AI Model	**Deep Learning** • Long Short-Term Memory (LSTM) • Gated Recurrent Unit (GRU) • Transformer • Deep Reinforcement Learning (DQN/PPO) (Optional Extension for dynamic channel switching)
Possible Inputs	• WiFi channel utilization • Signal-to-Noise Ratio (SNR) • Received Signal Strength Indicator (RSSI) • Channel interference level • Number of connected devices • Packet loss rate • Throughput • End-to-end latency • Jitter • Bandwidth utilization • Channel occupancy time • Access Point (AP) density • Historical traffic patterns • Frequency band (2.4 GHz / 5 GHz / 6 GHz)
Novel Contributions	• Develop an **AI-driven WiFi channel selection framework** that predicts the optimal channel based on real-time wireless conditions. • Compare **LSTM, GRU, and Transformer** models for channel prediction accuracy. • Reduce channel interference, congestion, and packet loss while improving throughput and QoS. • Enable adaptive channel switching in dynamic wireless environments. • Compare the proposed approach with conventional WiFi channel selection algorithms and IEEE 802.11 default mechanisms.
Dataset	• **AWID (Aegean WiFi Intrusion Dataset)** (for WiFi traffic features) • **CRAWDAD Wireless Datasets • WIDE Wireless Trace Dataset • Cisco Wireless Network Trace Dataset** (where available) • Custom datasets generated using **Wireshark**, **NS-3**, **Mininet-WiFi**, **OMNeT++**, or real WiFi testbeds with multiple access points and varying interference levels.
Research Gap	Existing WiFi channel selection methods primarily rely on **static configurations**, simple heuristic algorithms, or periodic channel scanning, which cannot effectively adapt to dynamic wireless environments. Most AI-based studies focus on traffic prediction or interference classification rather than **real-time intelligent channel selection**. There is limited research comparing modern deep learning models for congestion-aware channel prediction across **WiFi 5, WiFi 6, and WiFi 7** networks.
Novelty	Proposes a **Deep Learning-based intelligent WiFi channel selection framework** that predicts the **least congested wireless channel** using real-time network conditions instead of relying on fixed or heuristic channel assignments. The framework supports **adaptive channel optimization**, **interference mitigation**, and **QoS-aware wireless communication**. The novelty can be further enhanced by integrating **Deep Reinforcement Learning** for autonomous channel switching, **Graph Neural Networks** for multi-access-point coordination, **Federated Learning** for privacy-preserving optimization, or **Explainable AI (XAI)** to justify channel selection decisions.

Field	Details
Title 19	**Reinforcement Learning-Based Intelligent Wireless Handoff Prediction for Seamless Access Point Selection**
Problem Statement	Wireless devices traditionally perform handoffs only after the signal strength of the current **Access Point (AP)** drops below a predefined threshold. This reactive approach often results in **handoff delays, packet loss, increased latency, connection interruptions, and poor Quality of Service (quality of service)**, particularly for real-time applications such as video conferencing, VoIP, online gaming, and autonomous systems. Existing handoff mechanisms primarily rely on **RSSI-based threshold algorithms**, which cannot accurately predict user mobility or future network conditions. This research proposes a **Reinforcement Learning (RL)-based intelligent wireless handoff framework** that predicts the **best Access Point before the user moves**, enabling proactive handoff decisions that minimize service disruption and optimize network performance.
AI Model	**Deep Reinforcement Learning (DRL)** • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Soft Actor-Critic (SAC) (Optional Extension)
Possible Inputs	• Received Signal Strength Indicator (RSSI) • Signal-to-Noise Ratio (SNR) • User location • User mobility speed and direction • Access Point load • Available bandwidth • Channel utilization • End-to-end delay • Packet loss rate • Throughput • Handoff history • AP coverage area • Network congestion level • QoS requirements
Novel Contributions	• Develop a **proactive Reinforcement Learning-based wireless handoff framework**. • Predict the optimal Access Point **before signal degradation occurs**, enabling seamless connectivity. • Design a reward function that minimizes handoff latency, packet loss, and unnecessary handoffs (ping-pong effect). • Compare multiple DRL algorithms (DQN, PPO, A2C) for handoff optimization. • Evaluate performance across WiFi 6/6E/7, campus networks, enterprise WLANs, and mobile environments.
Dataset	**• CRAWDAD Wireless Mobility Datasets • WIDE Wireless Trace Dataset • Microsoft GeoLife Mobility Dataset** (for mobility prediction) **• Rome Taxi Mobility Dataset** (optional for mobility modeling) **• NS-3 Wireless Simulation Datasets** • Custom datasets generated using **Mininet-WiFi, NS-3, OMNeT++, Wireshark**, and real WiFi testbeds with user mobility scenarios.
Research Gap	Existing wireless handoff mechanisms mainly rely on **signal-strength thresholds** and reactive decision-making, leading to delayed handoffs and reduced QoS. Current AI-based approaches often optimize only a single metric such as RSSI or latency, while ignoring **user mobility, AP load, congestion, and future network conditions**. There is limited research on applying **Deep Reinforcement Learning** for **proactive Access Point prediction**, enabling seamless and autonomous handoff decisions in next-generation wireless networks.
Novelty	Proposes a **Deep Reinforcement Learning-based proactive wireless handoff framework** that predicts the **best Access Point before user movement causes signal degradation**. Unlike conventional RSSI-based handoff mechanisms, the proposed framework considers **mobility prediction, AP load, network congestion, QoS requirements, and traffic conditions** to achieve intelligent, seamless connectivity. Novelty can be further enhanced by integrating **Graph Neural Networks** for network topology awareness, **Federated Reinforcement Learning** for distributed learning, **Digital Twins** for network simulation, or **Explainable AI (XAI)** to justify handoff decisions.

Field	Details
Title 20	**AI-Based Bluetooth Interference Prediction for Proactive Wireless Communication Optimization**
Problem Statement	Bluetooth communication operates in the crowded **2.4 GHz ISM band**, where it coexists with **WiFi, ZigBee, microwave ovens, and other wireless devices**. As a result, Bluetooth networks frequently experience **interference, packet collisions, retransmissions, increased latency, reduced throughput, and degraded Quality of Service (QoS)**. Existing Bluetooth devices primarily rely on **Adaptive Frequency Hopping (AFH)**, which reacts to interference only after it has occurred. This reactive approach cannot prevent performance degradation in highly dynamic wireless environments. This research proposes an **AI-based Bluetooth interference prediction framework** that analyzes wireless channel conditions and **predicts interference before transmission**, enabling proactive channel selection and transmission scheduling to improve communication reliability and efficiency.
AI Model	**Machine Learning & Deep Learning** • Long Short-Term Memory (LSTM) • Gated Recurrent Unit (GRU) • Transformer • XGBoost • Deep Reinforcement Learning (Optional Extension for adaptive channel hopping)
Possible Inputs	• Bluetooth channel occupancy • RSSI (Received Signal Strength Indicator) • Signal-to-Noise Ratio (SNR) • Frequency hopping sequence • Packet Error Rate (PER) • Bit Error Rate (BER) • Packet retransmission count • WiFi channel utilization • ZigBee activity • Spectrum occupancy • Distance between devices • Device mobility • Throughput • Historical interference patterns
Novel Contributions	• Develop an **AI-driven Bluetooth interference prediction framework** that forecasts interference before packet transmission. • Compare LSTM, GRU, Transformer, and XGBoost models for interference prediction accuracy. • Integrate predictive interference information with **Adaptive Frequency Hopping (AFH)** to proactively avoid congested channels. • Reduce packet retransmissions, latency, and energy consumption while improving throughput and QoS. • Evaluate performance in environments with WiFi, ZigBee, and other coexisting wireless technologies.
Dataset	• CRAWDAD Wireless Trace Datasets • IEEE DataPort wireless spectrum datasets • WIDE Wireless Trace Dataset • Bluetooth traffic traces collected using **Ubertooth One**, **Wireshark**, or Bluetooth sniffers • NS-3 Bluetooth simulation datasets • Custom datasets generated using **Mininet-WiFi, NS-3, OMNeT++, or real Bluetooth testbeds** under varying interference scenarios.
Research Gap	Existing Bluetooth interference mitigation techniques primarily rely on **reactive Adaptive Frequency Hopping (AFH)** and threshold-based channel assessment, which respond only after interference has affected communication. Most AI-based wireless studies focus on WiFi or cellular networks, with relatively limited research on **predictive Bluetooth interference analysis**. There is a need for intelligent models capable of **forecasting interference before transmission**, enabling proactive spectrum management and improved wireless reliability.
Novelty	Proposes an **AI-based proactive Bluetooth interference prediction framework** that forecasts wireless interference **before packet transmission**, enabling intelligent channel selection and adaptive frequency hopping. Unlike conventional reactive interference avoidance techniques, the proposed framework leverages **deep learning and predictive analytics** to improve communication reliability, reduce retransmissions, and optimize QoS. The novelty can be further enhanced by integrating **Graph Neural Networks** for multi-device interference modeling, **Federated Learning** for privacy-preserving distributed learning, **Transformer-based temporal analysis**, or **Explainable AI (XAI)** for interpretable interference prediction.

Field	Details
Title 21	**Deep Reinforcement Learning-Based Intelligent Packet Scheduler for QoS-Aware Network Traffic Management**
Problem Statement	Traditional packet scheduling algorithms such as **First-In-First-Out (FIFO)**, **Fair Queueing (FQ)**, **Weighted Fair Queueing (WFQ)**, and **Priority Queueing (PQ)** use predefined scheduling policies that cannot dynamically adapt to changing traffic patterns or Quality of Service (QoS) requirements. In modern networks, including **5G/6G, Cloud Computing, Data Centers, and IoT**, these static scheduling techniques often result in **high end-to-end delay, packet loss, jitter, and inefficient bandwidth utilization** during periods of congestion. This research proposes a **Deep Reinforcement Learning (DRL)-based intelligent packet scheduler** that continuously learns network conditions and dynamically determines the optimal packet transmission order to minimize delay, packet loss, and jitter while maximizing throughput and fairness.
AI Model	**Deep Reinforcement Learning (DRL)** • Deep Q-Network (DQN) • Proximal Policy Optimization (PPO) • Advantage Actor-Critic (A2C/A3C) • Soft Actor-Critic (SAC) (Optional Extension)
Possible Inputs	• Queue length • Packet arrival rate • Packet priority • Packet size • Waiting time in queue • End-to-end delay • Jitter • Packet loss rate • Available bandwidth • Link utilization • Congestion level • Traffic class (VoIP, Video, Web, IoT, FTP, etc.) • Historical scheduling decisions • QoS requirements
Novel Contributions	• Develop a **Deep Reinforcement Learning-based packet scheduling framework** that replaces conventional FIFO and Fair Queueing algorithms. • Design a reward function that simultaneously minimizes **delay, packet loss, and jitter**, while maximizing throughput and fairness. • Enable adaptive scheduling under varying traffic loads and congestion conditions. • Compare DRL algorithms (DQN, PPO, A2C) with traditional scheduling methods such as FIFO, WFQ, PQ, and Deficit Round Robin (DRR). • Evaluate the scheduler across real-time applications such as VoIP, video streaming, IoT, and cloud traffic.
Dataset	**• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Internet2 Traffic Dataset • GEANT Traffic Dataset • SNDlib Network Dataset** • NS-3 generated scheduling datasets • Custom datasets generated using **NS-3, Mininet, OMNeT++, OpenFlow/SDN**, and cloud simulation environments under different traffic scenarios.
Research Gap	Existing packet scheduling algorithms rely on **fixed scheduling policies** and cannot adapt to dynamic network traffic or changing QoS requirements. Most AI-based scheduling studies optimize only a single metric, such as throughput or latency, while ignoring **multi-objective optimization** involving delay, packet loss, jitter, fairness, and bandwidth utilization. There is limited research on applying **Deep Reinforcement Learning** for **adaptive packet scheduling** in heterogeneous communication networks.
Novelty	Proposes a **Deep Reinforcement Learning-based intelligent packet scheduler** that replaces traditional **FIFO** and **Fair Queueing** with an adaptive, self-learning scheduling framework. Unlike conventional scheduling algorithms, the proposed approach continuously learns traffic behavior and dynamically prioritizes packets to optimize **delay, packet loss, jitter, throughput, and fairness**. The novelty can be further enhanced by integrating **Multi-Agent Reinforcement Learning**, **Graph Neural Networks**, **Digital Twins**, **Explainable AI (XAI)**, or **Intent-Based Networking (IBN)** for autonomous traffic management.

Field	Details
Title 22	**AI-Based Video Quality of Service Prediction Using LSTM and Transformer for Intelligent Multimedia Networks**
Problem Statement	Modern multimedia applications such as **video streaming, video conferencing, online gaming, cloud media services, and IPTV** require consistently high **Quality of Service (QoS)** to ensure a satisfactory user experience. Traditional QoS monitoring techniques detect quality degradation only after it occurs, leading to video buffering, increased latency, packet loss, and reduced Quality of Experience (QoE). Existing prediction methods often rely on statistical models that cannot accurately capture complex temporal network behavior. This research proposes an **AI-based Video QoS Prediction framework** that uses **LSTM** and **Transformer** models to forecast future video quality, enabling proactive network optimization, adaptive bitrate selection, and congestion avoidance.
AI Model	**Deep Learning** • Long Short-Term Memory (LSTM) • Transformer (Primary Comparison Model) • Gated Recurrent Unit (GRU) (Optional Baseline)
Possible Inputs	• Network throughput • End-to-end delay • Packet loss rate • Jitter • Available bandwidth • Video bitrate • Frame rate (FPS) • Resolution • Buffer occupancy • Startup delay • Rebuffering events • Round Trip Time (RTT) • TCP/QUIC statistics • Historical QoS and QoE measurements
Novel Contributions	• Develop an **AI-driven Video QoS prediction framework** using sequential deep learning models. • Compare **LSTM** and **Transformer** architectures for short-term and long-term QoS prediction. • Predict future video quality before degradation occurs, enabling proactive bitrate adaptation and traffic management. • Improve streaming performance by reducing buffering, latency, and packet loss. • Benchmark the proposed approach against traditional QoS prediction techniques and statistical forecasting models.
Dataset	**• LIVE-NFLX Video QoE Dataset • Waterloo Streaming QoE Database • LIVE Video Quality Database • YouTube UGC Video Dataset** (for QoE studies) **• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive •** Custom datasets generated using **NS-3**, **Mininet**, **OMNeT++**, **GStreamer**, or **Dash.js** under different network conditions.
Research Gap	Existing video quality prediction methods primarily rely on **reactive QoS monitoring** or traditional machine learning techniques, which are unable to accurately model long-term temporal dependencies in dynamic network environments. Most studies focus on either **network QoS** or **user QoE** independently, with limited research comparing **LSTM** and **Transformer** models for proactive video quality prediction across diverse multimedia applications.
Novelty	Proposes a **deep learning-based proactive Video QoS prediction framework** that forecasts future streaming quality using **LSTM** and **Transformer** architectures. Unlike conventional reactive monitoring systems, the proposed framework predicts QoS degradation before it impacts users, enabling **adaptive bitrate selection**, **intelligent traffic engineering**, and **QoE-aware multimedia delivery**. The novelty can be further enhanced by integrating **Reinforcement Learning** for adaptive bitrate control, **Graph Neural Networks** for CDN-aware optimization, **Federated Learning** for privacy-preserving QoE prediction, or **Explainable AI (XAI)** for interpretable QoS forecasting.

Field	Details
Title 23	**Deep Learning-Based Dynamic Bandwidth Allocation for Intelligent Network Resource Management**
Problem Statement	Traditional bandwidth allocation mechanisms assign **fixed or rule-based bandwidth** to users, applications, or network flows. These static approaches cannot effectively adapt to changing traffic demands, leading to **bandwidth underutilization, congestion, increased latency, packet loss, and degraded Quality of Service (QoS)**. Modern communication networks such as **5G/6G, Cloud Computing, Data Centers, SDN, and IoT** require intelligent bandwidth allocation strategies capable of responding to real-time network conditions. This research proposes a **Deep Learning-based Dynamic Bandwidth Allocation framework** that continuously learns traffic patterns and automatically allocates optimal bandwidth to maximize network utilization while minimizing congestion and QoS degradation.
AI Model	**Deep Learning** • Long Short-Term Memory (LSTM) • Transformer • Gated Recurrent Unit (GRU) • Deep Reinforcement Learning (DQN/PPO) (Optional Extension for adaptive allocation)
Possible Inputs	• Current bandwidth utilization • Network throughput • Link utilization • Queue length • Packet arrival rate • End-to-end delay • Packet loss rate • Jitter • Flow priority • Traffic type (VoIP, Video, Web, IoT, FTP, etc.) • Number of active users • Historical traffic demand • QoS requirements
Novel Contributions	• Develop a **Deep Learning-based intelligent bandwidth allocation framework** for dynamic network environments. • Compare **LSTM, GRU, and Transformer** models for bandwidth demand prediction and allocation decisions. • Dynamically allocate bandwidth based on predicted traffic demand instead of static policies. • Improve throughput, reduce congestion, and enhance QoS across heterogeneous network applications. • Evaluate the proposed framework in SDN, cloud, and wireless networking environments.
Dataset	**• CAIDA Internet Traffic Dataset • MAWI Working Group Traffic Archive • Internet2 Traffic Dataset • GEANT Traffic Dataset • SNDlib Network Dataset • Abilene Network Traffic Dataset** • Custom datasets generated using **NS-3**, **Mininet**, **OMNeT++**, **OpenFlow/SDN**, or cloud simulation environments under varying traffic conditions.
Research Gap	Existing bandwidth allocation methods primarily rely on **static thresholds, heuristic algorithms, or predefined QoS policies**, making them ineffective in highly dynamic network environments. Many AI-based studies focus only on **traffic prediction** without integrating **intelligent bandwidth allocation**. There is limited research comparing modern deep learning architectures for **real-time dynamic bandwidth allocation** across diverse networking scenarios while optimizing multiple QoS metrics simultaneously.
Novelty	Proposes a **Deep Learning-based adaptive bandwidth allocation framework** that replaces **static bandwidth assignment** with intelligent, traffic-aware resource allocation. By leveraging **LSTM**, **GRU**, and **Transformer** models, the framework predicts future bandwidth demand and dynamically allocates network resources to optimize **throughput, latency, packet loss, and QoS**. The novelty can be further enhanced by integrating **Deep Reinforcement Learning** for closed-loop optimization, **Graph Neural Networks** for topology-aware allocation, **Federated Learning** for distributed bandwidth management, or **Digital Twins** for real-time network simulation.

Field	Details
Title 24	**AI-Based Intelligent IPv6 Address Management for Automated Address Allocation and Scalable Network Administration**
Problem Statement	IPv6 provides an enormous address space, enabling the growth of **IoT, Cloud Computing, 5G/6G, and enterprise networks**. However, managing IPv6 addresses manually or through static allocation policies can lead to **address conflicts, inefficient address utilization, administrative complexity, and poor scalability**, especially in large and dynamic environments. Existing address management systems (IPAM) primarily use rule-based mechanisms and lack the intelligence to optimize address allocation according to network growth and usage patterns. This research proposes an **AI-based IPv6 Address Management framework** that automatically assigns, optimizes, and manages IPv6 addresses to minimize address conflicts, improve scalability, and simplify network administration.
AI Model	**Machine Learning & Deep Learning** • XGBoost • Random Forest • Long Short-Term Memory (LSTM) (for address demand prediction) • Transformer (Optional Extension) • Reinforcement Learning (Optional for adaptive address allocation policies)
Possible Inputs	• Number of connected devices • Device type (IoT, Server, Mobile, PC, Sensor, etc.) • Network topology • Subnet information • IPv6 prefix allocation • Device mobility • Historical address utilization • DHCPv6 logs • SLAAC information • VLAN information • Network growth rate • Address allocation history • Device lifetime statistics
Novel Contributions	• Develop an **AI-driven IPv6 Address Management (AI-IPAM) framework** for automatic address assignment. • Predict future address demand and dynamically allocate IPv6 address blocks. • Reduce address conflicts and optimize prefix utilization. • Improve scalability for IoT, cloud, enterprise, and smart city networks. • Compare AI-based allocation with traditional DHCPv6 and SLAAC-based address assignment mechanisms.
Dataset	• RIPE Atlas IPv6 measurement datasets • CAIDA IPv6 traffic datasets • Internet2 IPv6 traffic traces • DHCPv6 server logs (enterprise environments) • NS-3 and OMNeT++ IPv6 simulation datasets • Custom datasets generated using **Mininet**, **NS-3**, **Cisco Packet Tracer**, **GNS3**, or enterprise IPv6 network simulations.
Research Gap	Existing IPv6 address management systems rely on **static IPAM policies, DHCPv6, or Stateless Address Auto-Configuration (SLAAC)** without incorporating intelligent decision-making. Current approaches optimize address allocation but do not predict future address demand or dynamically adapt to network expansion. There is limited research on integrating **AI-driven predictive allocation**, **automated prefix management**, and **adaptive IPv6 address optimization** for large-scale enterprise and IoT networks.
Novelty	Proposes an **AI-powered IPv6 Address Management (AI-IPAM) framework** that intelligently assigns and optimizes IPv6 addresses based on **network growth, device characteristics, and traffic behavior**. Unlike conventional DHCPv6 and SLAAC approaches, the proposed system performs **predictive address allocation**, **dynamic prefix optimization**, and **scalable IPv6 resource management**. The novelty can be further enhanced by integrating **Reinforcement Learning** for adaptive allocation policies, **Graph Neural Networks** for topology-aware address planning, **Digital Twins** for network simulation, or **Explainable AI (XAI)** for transparent address assignment decisions.

Field	Details
Title 25	**LLM-Based Intelligent Network Troubleshooting Assistant for Automated Log Analysis and Root Cause Diagnosis**
Problem Statement	Network administrators spend significant time manually analyzing **router logs, switch logs, firewall logs, server logs, and network monitoring data** to diagnose failures and resolve performance issues. Traditional troubleshooting tools rely on keyword matching, static rules, or expert knowledge, making fault diagnosis slow, inconsistent, and difficult in large-scale networks. With increasing complexity in **Cloud Computing, SDN, Data Centers, IoT, and 5G/6G networks**, there is a need for an intelligent assistant that can understand network logs, identify root causes, and recommend corrective actions. This research proposes an **LLM-based Network Troubleshooting Assistant** that accepts uploaded network logs, automatically diagnoses network problems, explains the root cause, and suggests practical troubleshooting steps in natural language.
AI Model	**Large Language Models (LLMs)** • Llama 3 • GPT-based LLMs • Mistral • Qwen • Retrieval-Augmented Generation (RAG) • Vector Database (FAISS/ChromaDB) for knowledge retrieval
Possible Inputs	• Router logs • Switch logs • Firewall logs • Syslog messages • SNMP alerts • Network monitoring logs • Interface statistics • Routing tables • BGP/OSPF logs • DNS/DHCP logs • Packet capture summaries (PCAP metadata) • Device configurations • Error messages • Performance metrics (CPU, memory, bandwidth, latency, packet loss)
Novel Contributions	• Develop an **LLM-based network troubleshooting assistant** capable of understanding heterogeneous network logs. • Automatically identify **root causes** and generate prioritized troubleshooting recommendations. • Integrate **Retrieval-Augmented Generation (RAG)** with networking knowledge bases, RFCs, and vendor documentation for accurate recommendations. • Generate natural-language explanations suitable for both novice and expert network administrators. • Compare the proposed LLM assistant with traditional rule-based troubleshooting systems and AI classifiers.
Dataset	• **LogHub Dataset** • **HDFS Log Dataset** (for log analysis benchmarking) • **BGL (Blue Gene/L) Log Dataset** • **Thunderbird System Logs** • **OpenStack Log Dataset** • **Cisco Syslog Samples** • **Graylog/Elastic Stack logs** • Custom datasets generated from **Cisco Packet Tracer**, **GNS3**, **Mininet**, **NS-3**, **OpenFlow/SDN controllers**, and enterprise network devices.
Research Gap	Existing network troubleshooting systems rely primarily on **manual analysis**, **rule-based expert systems**, or limited machine learning models that cannot understand the semantic context of complex network logs. While LLMs have demonstrated strong reasoning capabilities, there is limited research on applying **LLMs with Retrieval-Augmented Generation (RAG)** for **network-specific root cause analysis**, multi-vendor troubleshooting, and explainable fault diagnosis.
Novelty	Proposes an **LLM-powered intelligent network troubleshooting assistant** that performs **automatic log understanding, root cause analysis, and troubleshooting recommendation generation** using natural language reasoning. Unlike conventional monitoring systems, the proposed framework combines **LLMs**, **Retrieval-Augmented Generation (RAG)**, **network knowledge bases**, and **explainable AI** to deliver context-aware, human-readable troubleshooting guidance. Future extensions can incorporate **agentic AI**, **multi-modal log analysis**, **network digital twins**, **autonomous remediation**, and **vendor-specific configuration generation**.

Field	Details
Title 26	**LLM-Based Intelligent Cisco Packet Tracer Lab Generator for Automated Networking Education**
Problem Statement	Designing **Cisco Packet Tracer** laboratory exercises is time-consuming and requires instructors to create network topologies manually, assign IP addresses, configure routing protocols, develop troubleshooting scenarios, and prepare assessment materials. Existing educational tools provide limited automation and require significant instructor effort to create diverse, curriculum-aligned labs. This research proposes an **LLM-based Packet Tracer Lab Generator** that automatically generates complete networking laboratories from natural language prompts, including network topology, device configurations, IP addressing, routing protocols, VLANs, ACLs, firewall rules, QoS settings, and assessment questions. The proposed system aims to reduce instructor workload while providing personalized, scalable, and interactive networking education.
AI Model	**Large Language Models (LLMs)** • GPT-based LLMs • Llama 3 • Qwen • Mistral • Retrieval-Augmented Generation (RAG) • LangChain/LangGraph (Optional for workflow orchestration)
Possible Inputs	• Course name (CCNA, CCNP, Computer Networks, etc.) • Topic (Routing, Switching, VLAN, OSPF, BGP, IPv6, ACL, NAT, QoS, Wireless, etc.) • Learning objectives • Difficulty level (Beginner/Intermediate/Advanced) • Number of routers, switches, PCs, and servers • Network size • IP addressing requirements • Routing protocol selection • Security requirements • Assessment objectives • Time duration of lab
Novel Contributions	• Develop an **LLM-powered Packet Tracer Lab Generator** that automatically creates complete networking laboratories from natural language instructions. • Generate **network topologies**, **device configurations**, **IPv4/IPv6 addressing**, **routing protocols**, **VLANs**, **ACLs**, **NAT**, **QoS**, and **wireless network configurations**. • Automatically generate **student instructions**, **expected outputs**, **troubleshooting tasks**, **quiz questions**, and **grading rubrics**. • Support multiple networking curricula (CCNA, CCNP, university courses, and certification training). • Evaluate the educational effectiveness of AI-generated labs compared with manually created laboratories.
Dataset	• Cisco Networking Academy Packet Tracer sample labs (where licensing permits) • Public GitHub repositories containing Packet Tracer exercises • Network configuration repositories (Cisco IOS examples) • RFC documents and networking textbooks for knowledge retrieval • Custom dataset of instructor-created Packet Tracer labs collected from university networking courses.
Research Gap	Existing networking education platforms primarily provide **static laboratory exercises** that require manual creation and updating by instructors. Current LLM applications in education focus mainly on text generation or tutoring rather than **automated laboratory generation**. There is limited research on using **LLMs with networking domain knowledge** to generate **complete, executable Packet Tracer laboratories**, including topology design, configurations, troubleshooting exercises, and assessment materials tailored to learning outcomes.
Novelty	Proposes an **LLM-based intelligent networking education platform** that automatically converts **natural language learning requirements** into **complete Cisco Packet Tracer laboratory exercises**. Unlike conventional educational tools, the proposed framework generates **network topologies**, **device configurations**, **assessment questions**, **troubleshooting scenarios**, and **grading rubrics** with minimal instructor intervention. The novelty can be further enhanced by integrating **multi-agent AI**, **Retrieval-Augmented Generation (RAG)**, **automatic Packet Tracer (.pkt) file generation**, **adaptive difficulty adjustment**, **student performance analytics**, or **digital twin-based lab validation**.

Field	Details
Title 27	**LLM-Based AI Network Configuration Generator for Automated Cisco Network Configuration**
Problem Statement	Configuring enterprise networks requires network administrators to manually write **Cisco IOS commands** for services such as **VLANs, DHCP, OSPF, ACLs, NAT, QoS, and IPv6**. Manual configuration is time-consuming, error-prone, and requires significant networking expertise. Existing configuration tools provide templates but cannot understand natural language requirements or automatically generate complete, optimized configurations. This research proposes an **LLM-based AI Network Configuration Generator** that accepts natural language requirements (e.g., *"I need VLAN + DHCP + OSPF"*) and automatically generates correct, optimized, and deployment-ready Cisco IOS configurations, along with configuration explanations and validation.
AI Model	**Large Language Models (LLMs)** • GPT-based LLMs • Llama 3 • Qwen • Mistral • Retrieval-Augmented Generation (RAG) • LangChain/LangGraph (Optional for workflow orchestration)
Possible Inputs	• Natural language network requirements • Number of VLANs • VLAN IDs and names • IP addressing scheme (IPv4/IPv6) • DHCP scope requirements • Routing protocol (OSPF, EIGRP, RIP, BGP) • ACL and firewall policies • NAT configuration • QoS requirements • Device type (Router, Switch, Layer-3 Switch, Firewall) • Enterprise topology details • Security policies
Novel Contributions	• Develop an **LLM-powered network configuration generator** that converts natural language into valid Cisco IOS configurations. • Automatically generate configurations for **VLANs, DHCP, OSPF, ACLs, NAT, QoS, IPv4/IPv6**, and other networking services. • Integrate **Retrieval-Augmented Generation (RAG)** with Cisco documentation, RFCs, and networking best practices for accurate configuration generation. • Provide **human-readable explanations** for every generated command to improve learning and trust. • Validate generated configurations using **Cisco Packet Tracer**, **GNS3**, **EVE-NG**, or **Cisco Modeling Labs (CML)** before deployment.
Dataset	• Cisco IOS configuration repositories (publicly available) • Cisco Validated Designs (CVDs) • RFC documents • Public GitHub repositories containing Cisco configuration examples • Network configuration datasets from GNS3, EVE-NG, or Packet Tracer labs • Custom dataset of enterprise network configurations curated from educational and laboratory environments.
Research Gap	Existing network automation tools primarily rely on **templates, scripts, or predefined playbooks**, requiring users to possess significant networking expertise. While LLMs have shown strong capabilities in code generation, there is limited research on **LLM-driven generation of production-ready Cisco IOS configurations** from natural language requirements. Furthermore, few systems provide **configuration validation, explainability, and automated compliance checking** within a unified framework.
Novelty	Proposes an **LLM-based intelligent network configuration generator** that translates **natural language networking requirements** into **validated Cisco IOS configurations**. Unlike conventional template-based automation tools, the proposed framework combines **LLMs**, **Retrieval-Augmented Generation (RAG)**, **networking knowledge bases**, and **configuration validation** to produce accurate, explainable, and deployment-ready network configurations. The novelty can be further enhanced by integrating **AI agents**, **automatic syntax verification**, **security policy compliance checking**, **multi-vendor configuration generation (Cisco, Juniper, MikroTik, Arista)**, **digital twin validation**, and **intent-based networking (IBN)**.